Stock Price Prediction Using Statistical and Machine Learning Techniques

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Abstract

Objectives: Stock price forecasting has always been a subject of research given its importance in the macroeconomics of a nation. It is tough to write down the future values of stocks using a specific set of equations. Numerous factors come into the picture when we talk about predicting the future price of a stock. Of all, the most important would be the historical price and volume data.

Methods: With the rise of machine learning, numerous approaches are proposed to forecast stock prices. Various models have been developed to date such as RNN, LSTM, CNN sliding window, etc., but were not accurate enough. This work is interested in predicting the price of a stock and comparing the results obtained using two algorithms namely Kalman Filters and XGBoost. Kalman filters are recursive in nature and use a feedback mechanism to perform error correction. This correction makes them best suited to make accurate predictions as they can factor in the market volatility, whereas XGBoost is a promising technique for datasets that are non-linear and can gather knowledge by detecting patterns and relationships in the data. XGBoost is additionally capable of capturing the time dependency of features efficiently.

Novelty: In the end, a hybrid model was developed by combining Kalman filters and XGBoost to give the perfect prediction needed to make future investments and to forecast stock. The average accuracy of XGBoost is higher when compared with Kalman filter and hybrid model.

Findings: The hybrid model seems to predict the future trend of the scripts with a good accuracy, but it is not able to do so for all the scripts. The average accuracy produced by the XGBoost model for the NSE dataset is 88.66%, and the NYSE dataset is 90.11%. The average accuracy produced by the Kalman Filter model for the NSE dataset is 89.09, and the NYSE dataset is 64.96. The average accuracy yielded by the hybrid model for the NSE dataset is 76.79%, and the NYSE dataset is 70.91%. However, for the individual stocks the hybrid model outperformed the XGBoost and Kalman filter.

Keywords: XGBoost, Kalman Filter, Hybrid Model, NSE, NYSE, Market Sentiment

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Cover Letter

From

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To

The Editor in-Chief,
.

Respected Editor in-Chief,

Subject: Cover Letter – Regarding.

I would like to submit my original article to **Indian Journal of Science and Technology**. with the title "**Stock Price Prediction Using Statistical and Machine Learning Techniques**". The co-author is **Dr. Venkata Vara Prasad** has contributed to the manuscript, and Dr.Prasad do agree to publish this article in the journal. We would like to publish our article in this journal because of the reason that this research work includes prediction of stock prices which comes under the scope of the journal.

Kindly please consider our manuscript for the peer-review process.

Thanking You

Yours Truly

Dr. Srinivas Gumparthi

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Stock price forecasting has always been a subject of research given its importance in the macroeconomics of a nation. It is tough to write down the future values of stocks using a specific set of equations. Numerous factors come into the picture when we talk about predicting the future price of a stock. Of all, the most important would be the historical price and volume data.

With the rise of machine learning, numerous approaches are proposed to forecast stock prices. Various models have been developed to date such as RNN, LSTM, CNN sliding window, etc., but were not accurate enough. Here, we are interested in predicting the price of a stock and comparing the results obtained using two algorithms namely Kalman Filters and XGBoost. Kalman filters are recursive in nature and use a feedback mechanism to perform error correction. This correction makes them best suited to make accurate predictions as they can factor in the market volatility, whereas XGBoost is a promising technique for datasets that are non-linear and can gather knowledge by detecting patterns and relationships in the data. XGBoost is additionally capable of capturing the time dependency of features efficiently.

In the end, a hybrid model was developed by combining Kalman filters and XGBoost to give the perfect prediction needed to make future investments and to forecast stock. The average accuracy of XGBoost is higher when compared with Kalman filter and hybrid model. The average accuracy yielded by the hybrid model for the NSE dataset is 76.79%, and the NYSE dataset is 70.91%.

The average accuracy produced by the Kalman Filter model for the NSE dataset is 89.09, and the NYSE dataset is 64.96. The hybrid model seems to predict the future trend of the scripts with a good accuracy, but it is not able to do so for all the scripts. The average accuracy produced by the hybrid model for the NSE dataset is 76.79, and the NYSE dataset is 70.91. However, for the individual stocks the hybrid model outperformed the XGBoost and Kalman filter.

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1. Introduction

Stock Market is an important investment field for many people nowadays. A common man is also interested in investing under stocks. The stock market prices are highly volatile subject to the environmental, political and other societal factors. So, an extensive analysis of stock price predictions is required. Earlier statistical models and machine learning models were used separately to predict the stock prices. But as the data becomes voluminous considering the historical data, the accuracy of these models in predicting stock prices have come down. So, in the current work, the authors have proposed a hybrid model which combines the statistical and machine learning model. The advantages of Kalman filter (statistical model) and XGBoost (machine learning model) was exploited to improve the prediction accuracy.

Many researchers had approached this problem using various Statistical and Machine Learning models. Few had even worked on a combination of these models. Very few gave significant results for the Indian Stock Markets. The research by Chatziset al.[1] aims at predicting the stock market crisis events using deep and statistical machine learning techniques. The approach is based on finding the probability of stock market crash events in various time frames. Cross-contagion effects among stock, bond, and currency markets were inspected. Daily returns were predicted using the models, and daily volatility of returns was computed by squaring log-returns. XGBoost had the best empirical performance on both 1-day and 20-day horizons.

Sen et al. [2] used ANN with a backpropagation algorithm for the training phase and a multilayer feed-forward network as a network model for predicting the price of a share. A decision support system using ANN for stock trading is studied. It also discusses previous works concerning Neural Network based on the decision-making process. The model varies for different combinations of input values and epochs. It outputs a performance curve, error curve, and output graph. A good validation performance and a regression value of 0.996 was obtained.

Dey et al. [3] compares various Deep Learning algorithms against XGBoost to predict stock market returns on Yahoo's dataset. The predictions were made for a period of 28-days, 60-days, and 90-days. Accuracy, Prediction, Recall, and Specificity has been calculated for each. The False Positive Rate is plotted for each of the models.

Karyaet al. [4] in their paper, Ensemble Kalman Filter Square Root method (EnKF-SR) and Ensemble Kalman Filter method (EnKF) were used to predict the stock price. The simulation results by the authors showed that the result estimation by applying the EnKF method was more precise than EnKF-SR, with an estimation error of about 0.2\% by EnKF compared to an estimation error of 2.6% by EnKF-SR.

Mortezaet al. [5] project tries to predict future stock prices using machine learning techniques on the NSE. They use linear regression and SVM regression. Linear regression will be used for predicting the opening price of the stock for the next day using the closing price of the stock for the previous day. SVM regression will be used for predicting the difference between the close and open price of the stock for the next day. External factors like foreign exchange rate, NSE index, moving averages, relative strength index, etc. are used to get maximum accuracy.

Dev Shah et al. [6], in his paper, discusses the technical and fundamental approaches for stock market analysis. Statistical, Machine Learning, Pattern Recognition, Sentiment Analysis, and Hybrid Techniques were considered in detail under Technical

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Analysis. It also considers Algorithmic Trading. The author concludes by stating that a combination of Hybrid and Statistical Machine Learning techniques will produce better results. Artificial Neural Network (ANN) is a part of Artificial Intelligence (AI), and it is a popular way to identify hidden and unknown patterns in data that is suitable for share market prediction. The past data of the selected stock is used for building and training the models.

Song et al. [7], in his paper compares RNN-LSTM model with SVM and XGBoost. The dataset was obtained from the API of Google Finance by applying the Python Library. Twenty companies traded at NASDAQ and NYSE were chosen. Indicators such as RSI, ADX and Parabolic SAR were considered. The results were compared by plotting the testing set errors.

The research by Wanjawaet al. [8] proposes the use of feedforward Artificial Neural Network with multi-layer perceptron using backpropagation to predict stock price. The model works in 4 stages in an iterative fashion. The dataset is obtained from the Nairobi Stock Exchange and NYSE. Tuning, by varying the number of hidden layers and perceptions. Each tuning experiment is done based on the model from the previous experiment. The final model is of a configuration of 5:21:21:1, using 80\% of the available data for training. Root Mean Square Error is used as a parameter to compare the results obtained with the original results. The model proved good for prediction of Stock Prices.

Xu et al. [9] performs a comparative literature survey to prove that the results of ANN are more accurate compared to SVM. Several Feed Forward ANNs are trained using Back Propagation. The assessment is made on the NASDAQ Stock Exchange, considering a 6-month dataset. The input to the model is short term historical stock prices and day of the week. The architecture of the neural network was optimized by applying different values for the number of neurons and the number of hidden layers in each hidden layer.

Hongying Zheng et al. [11] proposes a new deep learning model Random Long Short-Term Memory (RLSTM) which aims at preventing overfitting. The model is made up of a ModAugNet-c framework which augments the potentials of a single LSTM module by having - one LSTM module that can mitigate the risk of overfitting using additional reinforced data that has a strong correlation with the stock data that's considered for prediction, and another LSTM module for prediction. This was tested on SSEC and S&P500 datasets, and the model outperformed other models used for the dataset in terms of accuracy.

Widodo Budiharto [12] proposes a model which is a combination of LSTM and R Language based statistical computing to forecasts stocks of Indonesian exchange in the Covid-19 pandemic. The model performed well for short term data which are for predictions less than a year and gave an accuracy of 94.5%, over predictions for long term data.

Salvatore M. Cartaet al. [13] uses a Machine Learning approach to measure the magnitude of future stock price fluctuations with the help of binary classification techniques using Decision Trees. Lexicons are identified and generated from globally published articles, industry related news etc. These are fed to the prediction model

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along with the extracted features. This was tested on S&P 500 companies, but the accuracy was only 50-60%.

Milad et al. [14] compares the usage of artificial neural networks and heuristic algorithms such as Spider Optimization and Bat Algorithm with conventional time series models such as ARMA and ARIMA to predict stock prices. These models were used on various international indices such as Nasdaq, S&P500 and DJI. The ANN models exhibited the least errors and gave a better forecast when compared to the time series models.

Zelingheret al. [15] tests the usage of various Machine Learning models to anticipate the fluctuations in maize prices. They considered Linear models, Classification and Regression trees, Random Forest, Gradient Goosting. These were based on Root Mean Square Error (RMSE), Leave one out Cross Validation(LOOCV) etc and were proved effective for prediction of maize prices.

Chetan et al. [16] did a comparative study using several Machine Learning Algorithms such as KNN, Logistic Regression, Naïve Bayes, Random Forest, SVM, Decision Trees etc., combined with Sentiment Analysis to predict stock prices during the Covid-19 pandemic due the vast availability of sentiments in social media at this period. Among the models considered, the logistic regression model performed the best, whereas KNN model gave the least accuracy.

2. Design and methodology

The architecture diagram for the prediction model is shown in Figure1. The dataset obtained from NSE is first cleaned to check for null values and get with Market Sentiments data collected from Yahoo Finance. It is then pre-processed to obtain various other features such as Adjusted Close Factor, Adjusted Close Shift, Open Shift, Open Difference, Close Difference, High Difference and Low Difference. The input is taken from stock data which is being cleaned. This data is then sent to each of the three models. In the first model - The Kalman Filter, the data goes through Predict and Update steps at every iteration. In Predict step, Closing Price for the next time step, k is computed from the features being updated at the previous time step, k-1. In the Update step, the features are updated corresponding to the prediction at the last time step. In XGBoost, the dataset goes through two phases - Fit and Predict. In Fit phase, the model recursively builds various decision trees to narrow down at the one with least loss function value. The Predict phase uses this decision tree to make further predictions. For the Hybrid model we first run the XGBoost algorithm to perform a feature selection. These features are keyed as input to Kalman Filter which predicts the Closing price of the stock. We then perform an output analysis to compare the results of the model.

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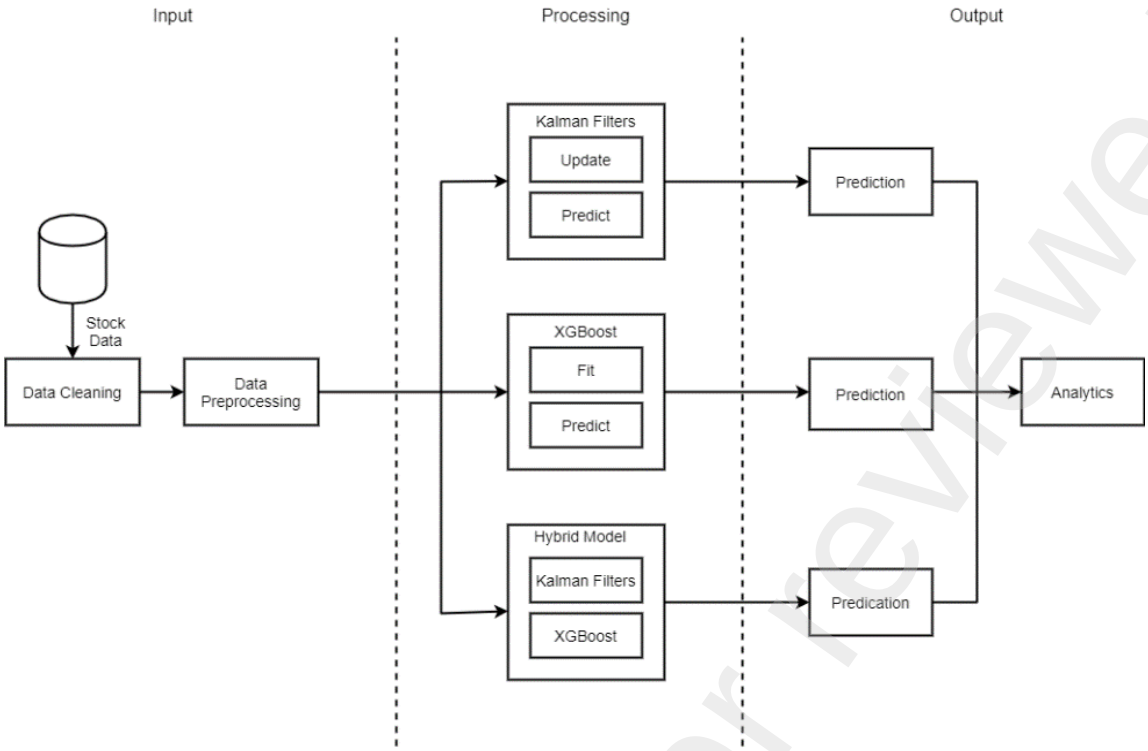


Figure 1: Architecture Diagram

2.1. Kalman Filter

In statistics, Kalman filtering is an algorithm that uses a series of measurements containing statistical noise and inaccuracies observed over time. The algorithm produces estimates of unknown variables that tend to be more accurate than those based on a single feature alone. It estimates the joint probability distribution of the variables for each timeframe. The Kalman filter is named after Rudolf E. Kalman, who is one of the primary developers of the theory.

$$X_k=F_k* X_k+ B_kU_k+ W_k$$

Where,

F_k is the state transition model which is applied to the previous state X_{k-1},

B_k is the control-input model, which is applied to the control vector U_k,

W_k is the process noise which is assumed to be drawn from a zero mean multivariate normal distribution.

2.2. XGBoost

XGBoost is an efficient and popular gradient boosted tree algorithm which is implemented in an open-source manner. Gradient boosting is a supervised learning algorithm. XGBoost attempts to accurately predict a target variable by combining the estimates of a set of simpler, weaker models.

XGBoost is an ensemble learning method. Ensemble learning offers a systematic solution that combines the predictive power of multiple learners. The resultant is a single model

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that has the aggregated output from several models. The base learners that form the ensemble could be either from the same learning algorithm or different learning algorithms. The widely used ensemble learners are bagging and boosting. The most predominant usage of XGBoost has been decision trees followed by statistical models.

When using gradient boosting for regression, each regression tree maps a data point to one of its leaf's which has a continuous score and, the weak learners are regression trees. XGBoost minimizes a regularized objective function that combines a convex loss function and a penalty term for model complexity. The training proceeds iteratively, adding new trees that predict the residuals of prior trees that are then combined with previous trees to make the final prediction. It's called gradient boosting because it uses a gradient descent algorithm to minimize the loss when adding new models.

3. Dataset and Implementation

3.1. Dataset

The dataset contains the stock values of 10 scripts from New York Stock Exchange (NYSE) and National Stock Exchange(NSE). The name of the stocks and their symbols are represented in the table Table1 given below.

Table 1 Table showing the datasets considered

NSE		NYSE	
Symbol	Company Name	Symbol	Company Name
HDFCBANK	HDFC Bank	APA	Apache Corporation
HINDUNILVR	Hindustan Unilever Limited	GE	General Electric Company
INFY	Infosys Limited	GPS	Gap Inc.,
ITC	ITC Limited	HPQ	Hewlett-Packard
LT	Larsen & Toubro Limited	JPM	JPMorgan Chase & Co.
RELIANCE	Reliance Industries Limited	LUV	Southwest Airlines Co.,
SUNPHARMA	Sun Pharmaceutical Industries Limited	MCD	McDonald's
TATAMOTORS	Tata Motors Limited	SWN	Southwestern Energy
TCS	Tata Consultancy Services Limited	VZ	Verizon Communications
WIPRO	Wipro Limited	YUM	Yum! Brands Inc

A sample of HDFCBANK data is shown in Table 2. It consists columns of Open, Close, High, Low, Volume traded, Adjusted Close, EPS, P/E Ratio. We have considered both the Fundamentals of the stock as well as Market Sentiments which helped us predict better.

Table 2 HDFC Bank Dataset from NSE

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	Date	Open	High	Low	Close	Adj Close	Volume	EPS	PE Ratio
2	03-01-2011	237	239.965	235.04	239.05	196.6809	5398390	34.3296	65.81
3	04-01-2011	238.65	238.8	233.3	234.555	192.9827	8284130	34.3296	65.81
4	05-01-2011	234.97	234.97	229.75	230.82	189.9096	7825920	34.3296	65.81
5	06-01-2011	231.6	234.265	230.9	232.775	191.5181	5435090	34.3296	65.81
6	07-01-2011	232.62	232.62	226.1	226.945	186.7215	7948010	34.3296	65.81
7	10-01-2011	225.605	227	213.8	215.225	177.0787	20070860	34.3296	65.81
8	11-01-2011	217	221.87	212.5	214.66	176.6138	12157590	34.3296	65.81
9	12-01-2011	216.9	221.4	210.66	220.53	181.4434	9905330	34.3296	65.81
10	13-01-2011	220.9	221.74	212.64	214.3	176.3176	9486010	34.3296	65.81
11	14-01-2011	215	220.8	203	204.955	168.6289	12791410	34.3296	65.81
12	17-01-2011	206	209.3	205.055	206.81	170.1551	12280820	34.3296	65.81
13	18-01-2011	208.68	211.6	204.665	210.935	173.549	8420160	34.3296	65.81

We found the data correlation between each pair of features using the Pearson's correlation matrix. This is depicted in Figure 2. Data Correlation helps to understand the relationship between multiple variables and attributes in the dataset. Using Correlation, we can get some insights such as one or multiple attributes depend on another attribute or a cause for another attribute and one or multiple attributes are associated with other attributes.

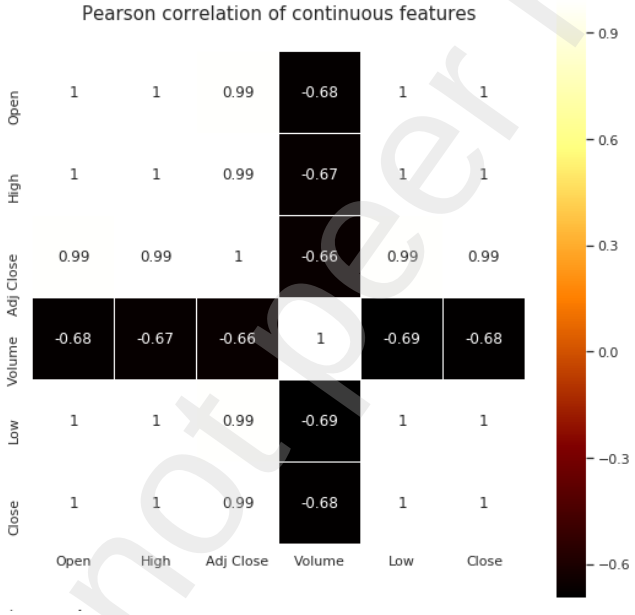


Figure 2 Correlation of Features from the dataset

The dataset is pre-processed to obtain other features such as Adjusted Factor, Adjusted Close Shift, Open Shift, High Difference, Low Difference, Open Difference and Close Difference. The dataset after pre-processing in shown in Table 3.

Table 3 Pre-processed HDFC Dataset

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Date	Open	High	Low	Close	Adj Close	Volume	EPS	PE Ratio	Adj_factor	Adj Close	Open shif	High_diff	Low_diff	Close_diff	Open_diff
03-01-2011	194.9943	197.4338	193.3817	239.05	196.6809	6561313	34.33	65.81	0.8227607	192.9827	196.3519	4.451108	0.39901	3.698288	3.369206
04-01-2011	196.3519	196.4753	191.9501	234.555	192.9827	10068698	34.33	65.81	0.8227608	189.9096	193.3241	6.565657	2.040472	3.073029	3.414452
05-01-2011	193.3241	193.3241	189.0293	230.82	189.9096	9511782	34.33	65.81	0.8227607	191.5181	190.5514	1.805977	-2.48883	-1.60847	-0.96673
06-01-2011	190.5514	192.744	189.9754	232.775	191.5181	6605919	34.33	65.81	0.8227606	186.7215	191.3906	6.022563	3.25397	4.796646	4.669158
07-01-2011	191.3906	191.3906	186.0262	226.945	186.7215	9660171	34.33	65.81	0.8227608	177.0787	185.6189	14.31193	8.947543	9.642777	8.540248

The Adjusted Close Shift and Open Shift helps in relating the previous day's Adjusted close and Open with next day's Adjusted Close and Open. Each of Open, High, Low and Close are multiplied by Adjusted factor to make it correspond to Adjusted Close Price.

The results of the study conducted by Dr. Pankaj Kumar[11] says that earnings per share have found to be a very solid forecaster of the market price of a share of select companies and price-earnings ratio impact significantly on the prediction of the market price of a share. So overall, earnings per share is the main leading reflector in the performance of market price of a share.

3.2. Kalman Filter

3.2.1. Initialization

A Kalman Filter class is defined initially with data members F - the state transition applied to the previous step, H - which maps the right states to observed states, Q, R and P - the noise covariance matrices and posteriori state estimate - initialized to identity matrix and x - the mean or predicted state estimate.

3.2.2. Prediction

This step must predict X the mean and P the covariance of the system. The function predict performs the prediction when given the Kalman object as input.

3.2.3Update

This step computes X, the mean, and P, the covariance of the system given the feature z to be updated and its standard deviation r. The function update performs the updation of X, y - the difference in measured and actual value, S, K - the Kalman Gain and P - the covariance. The matrix multiplications make use of dot product of the vectors.

For each time step k, perform the predict and update step to obtain the predicted values. This predicted array consists of the close price corresponding to the days for which the prediction must be made. These values depend on parameters such as Open, High, Low, Adjusted Close, Adjusted Close Shift, Open Shift, EPS, PE Ratio. After this step, we must calculate the error in the prediction by checking it against the Ground Truth. For this calculation, we use the mean absolute percentage error, which is calculated as error = [(measured – predicted)/(measured)] * 100. The graph for the prediction is plotted for the entire period under consideration. For this, we make use of the matplotlib package available in Python.

3.3. XGBoost

The algorithm involved and the implementation details of the XGBoost algorithm are discussed below in detail. The implementation is done in python using anaconda framework.

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3.3.1. Bagging

The decision trees exhibit highly variable behaviour, though decision trees are one of the most easily interpretable models. Consider a single training dataset that will be randomly split into two parts. Each part will train a decision tree to obtain two models. When it fits both these models, they will produce different results. Decision trees are said to be linked with high variance due to this behaviour. Bagging or boosting aggregation helps to decrease the variance in any learner. Several decision trees which are created in parallel, form the base learners of the bagging technique. Data sampled with replacement is given to these learners for training. The final prediction output is the mean output from all the learners.

3.3.2. Boosting

In boosting, all the trees are built sequentially, such that each succeeding tree intentions to reduce the errors of the preceding tree. Each tree learns from its antecedents and updates the residual errors. Hence, the tree all the succeeding trees will learn from an updated version of the residuals. The base learners whose bias is very high are weak learners, and the predictive power is just a little better than accidental guessing. Each of these weak learners contributes vital information for prediction, resulting in the production of a strong learner by combining these weak learners. The last and final strong learner has low bias and variance.

In stark contrast to bagging techniques like Random Forest, where trees are grown to their maximum extent, the boosting method in XGBoost makes use of trees with fewer splits. Such small trees are highly interpretable because the depth of the tree is very less. Parameters like the number of iterations or trees, the learning rate of gradient boosting, and the tree depth, could be optimally selected through validation techniques like k-fold cross-validation. Overfitting is caused by having many trees. So, it is necessary to choose the stopping criteria for boosting carefully.

3.4. Hybrid Model

The Hybrid model is a combination of the statistical model, Kalman filter and the machine learning model, XGBoost. Hybrid model aids in proper prediction of the stock values with increased accuracy and consistency across various datasets. Kalman filters are optimally used to estimate the variables of interests when they can't be measured directly, but an indirect measurement is available. For this indirect measurement fourteen parameters can be used. Using Feature Extraction, XGBoost Algorithm takes the top five parameters according to the weighted values among fourteen parameters. The top five parameters vary for different stocks. These five parameters are fed into Kalman filter, and the stock value is predicted for the required future date, and output is extracted.

Each of the below models were tested with a test-set consisting of 400 days stock data by keying it with the no. of days after which the prediction must be made. All the models were tested with two different datasets - one consisting of Price Movements and other consisting of both Price Movements and Market Sentiments. Price Movements consist of data such as Open, High, Low, Close, Volume, Dividend Yield, Turnover etc. Market Sentiments refer to the overall psychology of investors towards a particular security. This is the feeling of a general crowd which is revealed through price movements of securities traded. A very high P/E

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indicates that the given stock is priced high and could drop at any time. In such a case, it's said that the stocks are overbought. A low P/E indicates lack of confidence in the performance and future of a company. EPS denotes the earnings of a company per each share it holds. It is a measure of profitability of a company. A high EPS is usually preferred.

4.1. Kalman Filter

Below figures depict the model's HDFC Bank dataset predictions before and after considering Market Sentiments. We can see that the model is performing better for the latter case. With just the price movements we get an accuracy of only 88% as shown in Figure 3. But after including Market Sentiments we get an accuracy of 91.75% as depicted in Figure 4. The accuracy has improved greatly.

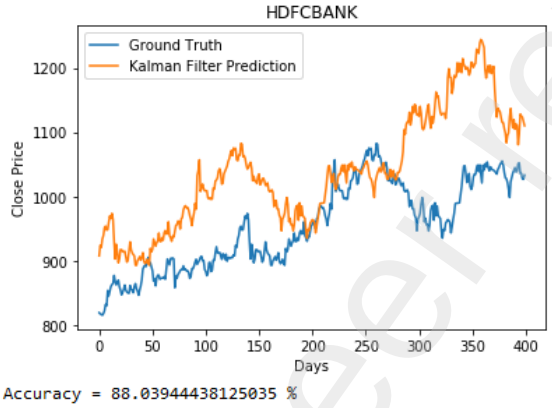


Figure 3 Predictions for HDFC without considering Market Sentiments

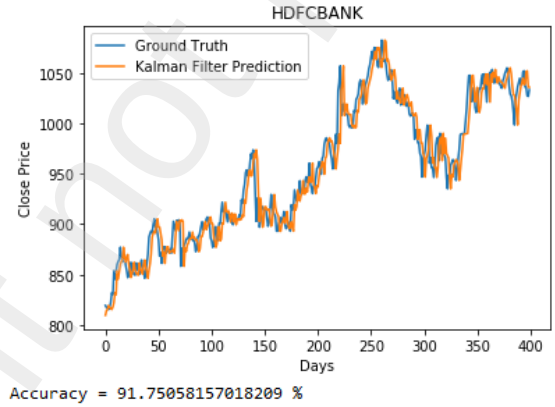


Figure 4 Predictions for HDFC considering Market Sentiments

The model was also tested for the rest of the 18 datasets available which are summarised in the Table 4 shown below.

Table 4 Accuracy of the Kalman filter model for different datasets

NSE		NYSE	
Symbol	Accuracy	Symbol	Accuracy
HDFCBANK	92.32	APA	92.63
HINDUNILVR	85.31	GE	45.40
INFY	92.03	GPS	38.05

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ITC	87.35	HPQ	85.18
LT	89.09	JPM	93.54
RELIANCE	85.31	LUV	77.39
SUNPHARMA	90.92	MCD	31.68
TATAMOTORS	90.37	SWN	68.56
TCS	85.57	VZ	76.20
WIPRO	92.68	YUM	41.00

From the above results we can see that the model performs consistently for various datasets from the NSE and can be relied on to make better predictions, but for NYSE the accuracies are not consistent and high.

4.2. XGBoost

The XGBoost model was developed for the HDFC Bank dataset from National Stock Exchange and JPMorgan and Chase dataset from New York Stock Exchange. The models were trained with and without market sentiments. Unlike Kalman Filters which produced significant changes in the accuracy and error rate after adding the market sentiments, the accuracy and error rate remained the same in XGBoost. So, the market sentiments were considered in further training. After building the models for the data with gave an accuracy of 91% and 88% for the testing data of JPMorgan Chase and HDFC Bank.

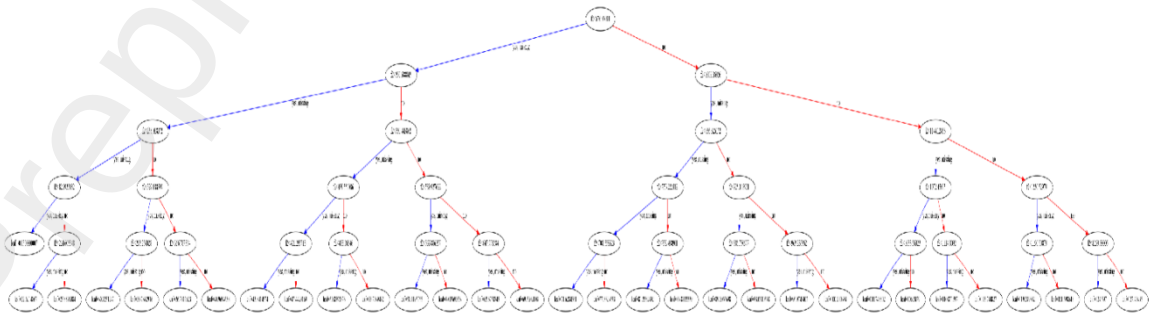
```
Mean Absolute Error     : 49.84193002359586
Mean Squared Error      : 3372.8825938149794
Root Mean Square Error : 58.076524
ACCURACY OF XGBoost Model: 88.36505213427611
```

Figure 5 Accuracy for HDFC Dataset considering market sentiments

```
Mean Absolute Error     : 3.77279453737262
Mean Squared Error      : 18.94811052258092
Root Mean Square Error : 4.352943
ACCURACY OF XGBoost Model: 91.18795288170085
```

Figure 6 Accuracy for JPM Dataset considering market sentiments

The Figure 5and Figure 6 shows the accuracy and errors obtained while testing the datasets of HDFC Bank and JPMorgan and Chase respectively.



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Figure 7Weights of the Boosted Regression Tree while training HDFC dataset

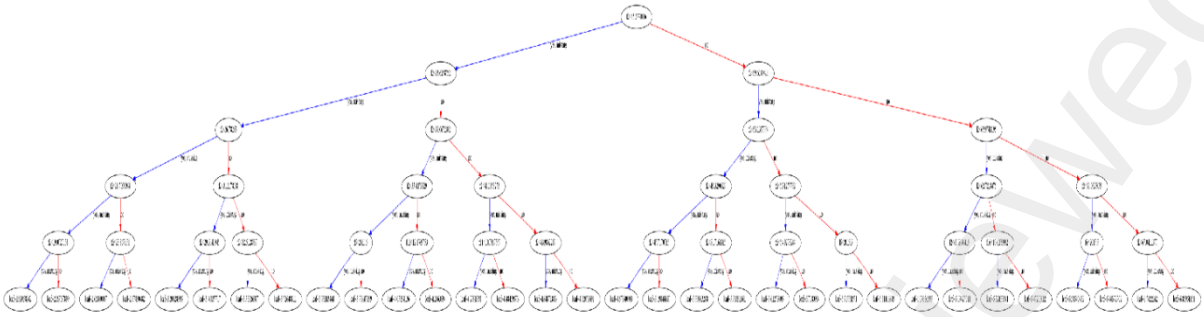


Figure 8 Weights of the Boosted Regression Tree while training JPM dataset

The Figure 7 and Figure8 shows the weights of the boosted regression trees obtained while training model which is then used for further predicting the training set.

The model was also tested for the rest of the 18 datasets available which are summarised in the Table 5 shown below.

Table 5 Accuracy of the XGBoost model for different datasets

NSE		NYSE	
Symbol	Accuracy	Symbol	Accuracy
HDFCBANK	88.04	APA	91.88
HINDUNILVR	88.08	GE	90.87
INFY	82.33	GPS	89.05
ITC	90.64	HPQ	92.81
LT	89.76	JPM	91.18
RELIANCE	94.16	LUV	86.23
SUNPHARMA	86.68	MCD	92.45
TATAMOTORS	91.80	SWN	90.28
TCS	86.20	VZ	89.47
WIPRO	88.95	YUM	86.87

4.3 Hybrid Model

The results of the model for two different company's stocks are shown below.

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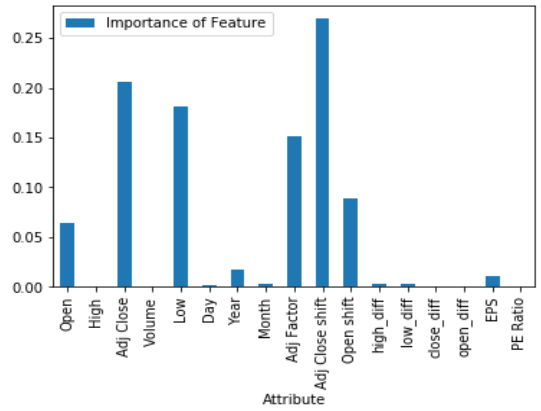


Figure 9 Feature importance found by XGBoost for HDFC dataset

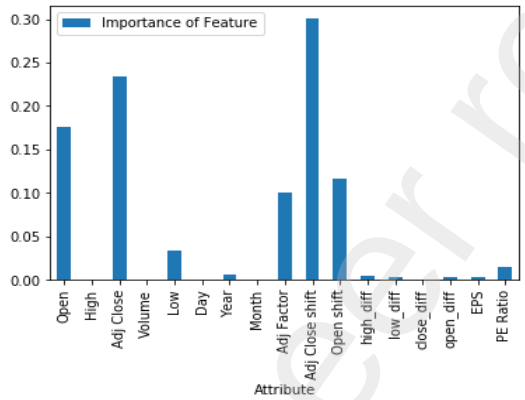


Figure 10 Feature importance found by XGBoost for JPM dataset

From the above Figures 9 and Figure10 obtained by implementing the XGBoost algorithm, we can find the features which play the highest importance in the boosted regression trees while prediction. We consider only the top five features from the above step and feed those features to the Kalman Filter model for the final stock value prediction.

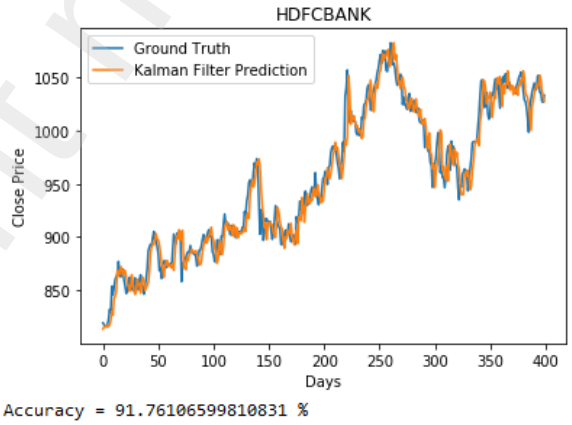


Figure 11 Predictions for HDFC Dataset by the Hybrid Model

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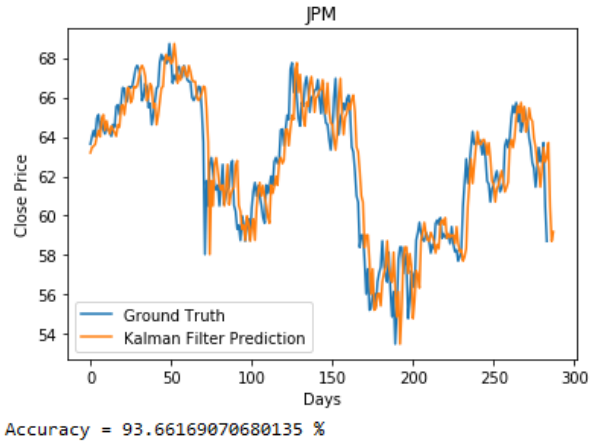


Figure 12 Predictions for JPM Dataset by the Hybrid Model

The model performs well consistently for different datasets. The Figure 11 and Figure 12 shows the accuracy of the model is better than those obtained for the other two individual models for both the datasets.

The model was also tested for the rest of the 18 datasets available which are summarised in the Table 6 shown below.

Table 6 Accuracy of the hybrid model for different datasets

NSE		NYSE	
Symbol	Accuracy	Symbol	Accuracy
HDFCBANK	87.57	APA	93.05
HINDUNILVR	77.16	GE	90.61
INFY	35.82	GPS	82.02
ITC	93.50	HPQ	93.84
LT	84.465	JPM	71.96
RELIANCE	92.70	LUV	43.08
SUNPHARMA	88.11	MCD	91.17
TATAMOTORS	23.12	SWN	56.24
TCS	93.87	VZ	36.04
WIPRO	91.62	YUM	51.12

4. Conclusion

The stock market prediction analysis was done on 20 scripts where 10 of them belonged to the National Stock Exchange (NSE), and the other 10 belonged to the New York Stock Exchange (NYSE). On average, the stock values for ten-year period were taken comprising of 2300 rows per script. The time interval considered where 2006 to 2016 and 2010 to 2020. The XGBoost model is consistent for all the twenty scripts taken for analysis. The average accuracy produced by the XGBoost model for the NSE dataset is 88.66, and the NYSE dataset is 90.11. Though the Kalman filter statistical model is producing impressive results, the accuracy of the model is not consistent. The average accuracy produced by the Kalman Filter model for the NSE dataset is 89.09, and the NYSE dataset is 64.96. The hybrid model seems to predict the future trend of the scripts with a good accuracy, but it is not able to do so for all the scripts.

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The average accuracy produced by the hybrid model for the NSE dataset is 76.79, and the NYSE dataset is 70.91. Adding the market sentiments for each quarter for all the scripts seems to increase the accuracy of the all the three models built. The hybrid model seems to outperform the other two models when the dataset consists of higher value stock data.

6. Future Works

All three models described in the thesis give good accuracy with consistent results. But the hybrid model came out as the best in future prediction of the stock values predicting the highs and lows of the market with increased efficiency. As future work for the project, we can consider a set of scripts to make script selection. Then we can perform portfolio management on these scripts where the model returns the investments required, acceptable risk and amount in hand for investment.

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